MINSEONG KIM

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Portfolio: minseong-kim.github.io

RESEARCH INTERESTS

Applied machine learning for manufacturing and semiconductor systems (anomaly detection, root-cause analysis, predictive maintenance); large language model (LLM) applications in regulated, safety-critical domains such as clinical research; reliable integration of AI into enterprise systems.

EDUCATION

Georgia Institute of Technology Atlanta, GA (Online)
M.S. in Computer Science — Admitted; enrollment beginning Spring 2027

Long Beach City College Long Beach, CA
A.S. in Electrical Technology – Automation Technician (in progress, expected June 2028) *08/2026 – Present*
- Full-time enrollment; coursework in electrical technology, calculus, and AI/machine learning.

California University of Management and Sciences Anaheim, CA
M.S. in Computer Information Systems — GPA 3.46/4.0 *01/2017 – 09/2019*
- Completed while working full-time and self-funding tuition and living expenses.

Jeonju University Jeonju, South Korea
B.E. in Information and Communication Engineering — GPA 3.99/4.5 (91.41/100) *03/1997 – 02/2006*
- Relevant coursework: Linear Algebra, Probability and Statistics, Calculus, Signals and Systems, Digital Communication, Data Structures, Object-Oriented Programming, Electromagnetics, Circuit Theory.
- Studies interrupted by mandatory military service in the Republic of Korea Air Force.

INDEPENDENT RESEARCH & TECHNICAL PROJECTS

Self-directed AI Portfolio 08/2025 – Present
Non-commercial prototypes built independently; not formal academic studies and not production deployments.
- **ClinicalMind – AI-Assisted Clinical Study Design.** https://minseong-kim.github.io/clinicalmind/ — LLM workflows for protocol drafting (arms, endpoints, eligibility), real-time CDISC coherence checking, and retrieval-augmented (RAG) search of similar protocols; immutable audit trail designed around ALCOA+ and 21 CFR Part 11 requirements.
- **MES Process Innovation Console (Solar).** https://minseong-kim.github.io/mes-pi-ai-console — Simulated ingot-to-module manufacturing; AI-assisted root-cause analysis and predictive maintenance using Random Forest and Isolation Forest models.
- **Battery MES·AI Integration Platform.** https://minseong-kim.github.io/battery-mes-ai-platform — Proof-of-concept models for defect classification, anomaly detection, predictive maintenance, and battery-life estimation; LLM-generated diagnostic explanations via Spring Boot REST APIs.
- **Switchgear MES·AI Bridge.** https://minseong-kim.github.io/switchgear-mes-ai-bridge — Six-module MES/AI console modeling FDC root-cause analysis, automated HiPot failure response, QMS re-inspection, and PLM/ERP engineering-change coordination.

PROFESSIONAL EXPERIENCE

Emet Mortgage Brea, CA
Full Stack Developer *11/2025 – 03/2026*
- Designed RESTful APIs for LendingQB integrations and built Node.js/Express services with React front ends.
- Developed and maintained enterprise mortgage systems integrating loan processing, document generation, and reporting workflows.
- Automated DocMagic document-generation workflows, including PDF preview and download.
- Designed loan-data processing logic, including agent mapping, lender/investor role transformation, and business-rule automation.
- Built reporting and data-extraction features for pipeline visibility; integrated Google Cloud and AWS storage for secure document management.

- Collaborated with operations, compliance, and business users to translate mortgage workflows into technical solutions.

FNS Inc Torrance, CA
IT Lead Manager (Full Stack Developer) *11/2021 – 08/2025*
- Led development and 24/7 production support of logistics applications (Ocean Tracking, Air Cargo Visibility, Proof of Delivery, mobile Transportation Management System).
- Integrated applications with ERP, WMS, TMS, customs, and third-party logistics systems via RESTful APIs.
- Developed mobile TMS features for new orders, track & trace, proof of delivery, empty return, and accident reporting.
- Provided Level 2/3 support with root-cause analysis of complex production issues; monitored performance and availability.
- Tuned SQL Server, Oracle, and MySQL databases; planned deployments, patches, and releases under change management.
- Led UAT and post-production verification, coordinated with vendors and business teams, and mentored team members.

Nmsi Inc Brea, CA
IT Lead Manager (Full Stack Developer) *06/2020 – 10/2021*
- Designed and supported mortgage origination, underwriting, and servicing applications (ASP.NET, .NET Core, C#, SQL Server).
- Built secure, service-oriented applications and integrated third-party systems via RESTful APIs and JSON/XML services.
- Administered Windows Server, IIS, and SQL Server environments; analyzed logs and tuned performance for reliability.
- Gathered requirements with loan officers and stakeholders; led UAT, deployments, and post-production support.
- Prepared technical specifications, data-flow diagrams, and deployment and troubleshooting documentation.

Very J Los Angeles, CA
IT Manager (Full Stack Developer) *11/2019 – 06/2020*
- Built a real-time inventory synchronization system between warehouses and retail stores.
- Designed an offline data synchronization system to keep operations running during network outages.
- Developed and supported Warehouse Management System (WMS) and Time Card applications.
- Administered Windows Server/IIS/SQL Server and integrated third-party systems via RESTful APIs and SQL.

Samsung Electronics Yongin (Giheung), South Korea
Project Leader (Software Engineer) *06/2011 – 12/2013*
- System LSI IP Automated Management System: led development of a platform for cataloging, searching, and reusing semiconductor design IP to shorten product development time.
- Built parametric and hierarchical search over IP libraries to maximize design reuse across product teams.
- Enterprise PLM: implemented Modular/Configurable BOM and Intelligent BOM so multiple end items could be created dynamically.
- Designed a JavaScript client library API that let SAP and legacy applications embed PLM screens.
- Built a Technical Document Management workspace, related-object search (parts, documents, projects, test plans, engineering changes), and a document-usage statistics dashboard.
- Ran proofs of concept on MVC and Flex frameworks; provided support to overseas users and trained PLM power users.

NTT DATA Fukuoka, Japan
Software Engineer *01/2007 – 09/2010*
- Developed core modules of an Aeronautical Information Publication (AIP) system; validated flight routes, navigation aids, and airspace data on the four-week AIRAC cycle for safety-critical operations.
- Implemented validation logic to detect errors in time-sensitive aeronautical data and protect data integrity.
- Contributed to code reviews, release planning, and performance improvements for large spatial and geographic datasets.
- Worked entirely in a Japanese-language professional environment.

Earlier Experience (South Korea)
- Daegu Bank – Project Leader, next-generation banking system (05/2016 – 12/2016)
- PV Technology Corporation – Project Leader, warehouse management system (12/2015 – 05/2016)
- Korea Post Information Center – Project Leader, smart electronic banking system (12/2013 – 06/2015)
- LG Hitachi (02/2011 – 06/2011), Nexon Mobile (09/2010 – 01/2011), Neobill (03/2006 – 01/2007), Moazio (09/2005 – 03/2006) – Software Engineer

HONORS

Academic Merit Scholarship, Jeonju University approx. 2001 – 2005
Awarded for multiple semesters based on academic performance.

TECHNICAL SKILLS

Languages: Python, Java, C#, JavaScript/TypeScript, SQL, Rust, Shell

AI/ML: scikit-learn models (Random Forest, Isolation Forest), LLM APIs, retrieval-augmented generation (RAG)

Systems: Spring Boot, .NET Core, Node.js, Django/FastAPI, React, Angular; Oracle, SQL Server, PostgreSQL, MySQL, MongoDB

Cloud/DevOps: AWS (EC2, RDS, S3, IAM), Google Cloud Run, Docker, Kubernetes, Jenkins, GitLab CI/CD

CERTIFICATIONS, SERVICE & LANGUAGES

Certifications: Oracle Certified Professional (OCP); Sun Certified Java Programmer (SCJP), Web Component Developer (SCWCD), Business Component Developer (SCBCD)

In Progress: California Electrician Trainee (ET) Registration (application submitted; pending issuance); OSHA safety training card (expected December 2026)

Military Service: Republic of Korea Air Force – completed full term of mandatory service

Languages: Korean (native); English (professional); Japanese (business professional, JLPT N1)